

Viet Dung Nguyen "Dynamic Prior Preference Learning for Scalable, Robust Deep Active Inference"

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hello welcome back now we have the much anticipated student presentation that your adviser Alex recently told us will be awesome by Viet dong win on Dynamic prior preference learning for scalable robust deep active inference so thank you V take it away hello uh hello hello everyone um my name is zet I'm from uh the neuro adap Computing Laboratory um Department in computer science seest technology um and I am uh Alex advis uh and today I'll be talking about Dynamic PR preference learning to scale Aus active INF so yeah first of all want to talk about the motivation for this so nowadays you know from fure work in robotics um excels in implementing robots behavior and planning for example um creating um we can you know creating autonomous robots that can actively explore in the envir um additionally um acquiring we can also do a lot of Robotics applications for example acquiring knowledge and learn skills continuously um and um this also envisioned um cognitive and developmental robotics um so there just a goal um for um robots robotics research um that's we can try to build a robots that can continuously develop through interactions with the environment and their learning processes um and it is also also better um to have interactions with their uh physical and social world in the manner of human learning and cognitive development so there's a lot of Robotics applications um around um and for example one application can be to build ASA robots and um building um social collaborative robots and stuff like that um so underlying um all these robot work um uh we it's like active inference research where we um often build a generative models um um it's so the development of Robotics actually depends a lot on activ framework because um there's a great need for research um that integrates the generative World model into robotics control system um as also deep learning is increasingly used in constructing complex action planning algorithms for solving control tasks um so um we can see that there's a need for scaling um active in system especially and deep active INF system um to integrate into the real world applications so um resar in active INF has led to many improvements um in terms of robustness of um model based agents that applies to both

M processes um and partially absorbable MVPs or P MVPs so an example of an mdp environment should be Mountain car um uh for those who um knows or studies reinforcement learning already and um this m carment environments um in mdp we only know POS velocity right um and that is the actual state of the environment um but when we get to PVP for example like robotic tasks um we as agents see only the image frame and we do not know the under lying states such as position and orientation of the block or um join state of the robots for example right um so basically active inference is basically a form of inference and learning that sticks to balance first of all the um goal orienting objectives or PRI preference um so this signal exploit um the instrumental signal um so that the agents can seek the goal um and try to pursue its goal um and the other thing it needs to balance the epistemic terms or epistemic signal where the foraging uh it's cares about foraging and exploration driving signals um so researcher often actually encounter great difficulty in providing the agent with effective prior preference as well as Computing useful instrumental signal from the agents the priors um especially in the real world applications where um the environment is very complex the robotic control task and um the trajectory or partway um to be desirable um and Tas specific goals um often requires uh a very very long um trajectory of preference um so it is not that ABS practical to provide uh a prior preference distributions especially the um stream or roll out of go images so that is the reason why we need to find a way to also learn to generate the prior preference signal in active inference so on the right here is just our Umar preliminary results um but we'll get to that in a moment so um yes so we actually uh given without uh all of this uh right we need to um uh we need to predict a u uh drawout trajectory of a goal um States dynamically right um it can be a goal state or goal observations but basically the goal is um if you look on the figure on top here um the top row is the real trajectory where you see the mountain car right there's a car that it needs to go backward and to build some kind of a momentum and then go full to the right to get to the um to the flag if it is only going to the right and it can it does not have the enough momentum to to go to to the top of the hill so um so in here uh is the um actual observations you can see that the trajectory is um a successful trajectory and the goal here is to um for each uh observation at each time step we predict a um immediate predict predicted PR preference um image or goal at that single time step so that we have um the instrumental signal or idea of uh what we are trying to learn and then um from that we learn the actions planning uh on uh from the PRI preference signal so in this talk we are we specifically analyze this issue um the issue of um uh prior preference learning um in the current literature um and discuss potential resolutions for scaling the instrumental signal

inference to active inference by introducing the um contrasted recurrent State prior preference or crsb um this is the model um learning framework that um we also have um done in the past so uh this methodology frames active inference agents in term of progressively constructing and adapt adapting um a prior preference um at each time step so this also facilitate the Dynamics emission of a useful or dense very dense um instrumental signal uh at each time step so we also um study how our prior preference adaptation scheme um when operating jointly with the Laten space driven epistemic foraging terms uh offers active inference agent um a very nice U flexibility in challenging problem con context um such as um spars to reward um environments um and also composed of uh continuous observation and action space right example um for example robotic task um they only we often like do not know the spars um do not know the dense reward um functions in the actual real environments and we we only know if the robot has risked the goal or not right um and also all these environments has you know varying goal States so um we can also see that providing the uh uh or manually craft the uh distributions of goal is very not very um practical so uh yeah so here here's the outline of this talk on the introduction I've had just covered um so we so first of all I'll just assume that's uh everyone is new to active inference and I'm just going to briefly cover active inference backgrounds and deep active inference um and also cover a little bit about um prior PRI preference learning and problems um uh of all those prior preference learning framework and then I'm just going now go through the actual methodology which is the dynamic Prime preference learning um and then first of all is the world model um next is the self-revision mechanism and then next is the um learning Prime preference models and then finally um putting all them together um to make an agent um and then um I'll show the results and then discussion okay so first of all um I want to um cover a little bit about active inference um so just a brief overview about how active inference work and what is the underlying theories behind it um so um it is a ma mathematical framework that describes how intelligent systems are human interact with the environment um so it describe it also describe the environment as the generative process um and the intelligent systems um are the agent or um us as generative model um so active INF is built on top of the free energy principles um this means that we as agent or intelligent system continuously observe the environment and forming our and correcting our belief or our own representation of the world um so the generative process um defines a real world environment here right um this is treated as the data Generation Um process where the information patterns are sensed by the agent so you can see that's the um hidden State underlying hidden States denoted as here um it's generate a sense of um some

kind of observations denoted o and this is also sensed by our intelligence system so on the right here is the generative model u_h is the actual agent that we are trying to build on the intelligent system right the intelligence system also has an internal State u_m which is trying to estimate the generative process States u_m so this is called the prior or the internal belief prior to the observations of the environment at a particular time stamp so this is denoted as the random variable s or state and P of s here u_m is the distribution over the random variable s so this makes the agent's generative model the generative model represents the agent's approximation of this generative process in this world by continuously seeing the observations or making the observation o_h and modified or updating u_h its belief so we actually call the belief u_m s after being updated after making the observation or the posterior because it is the posterior after making the observation right so in order to learn the model and estimate the posterior we first need to introduce these terms first of all is the prior over State u_m which is denoted as P_S which is also the distribution over state right u_m $P_{o|s}$ here is the model evidence u_m it is the distribution over the current observations denoting how well is our model at predicting the real data u_m and in order to compute this easily this is often written as the sum of probability u_h of observation given every single state and then next one is p of O given s this is the likelihoods u_m and the probability u_h this is the probability of an observation given a state so this is a function it gives a probability given an input States and then the last one is the posterior functions u_h p as given o which is the posterior function that Maps the observation to the states u_m so u_h normally in traditional active inference the prior function would be the transpose of the likelihood Matrix for example u_m and deep learning literature u_m it is often called an encoder and the likelihood here is a decoder so how do we estimate the posterior now u_m we have to find belief that maximizes the model evidence right because u_m at the current time points u_h there is only one single observation o here and it is real right so P of o p of that observation u_m it's going to be u_h one right so it should be the probability of one u_m and then u_m we use the state to estimate the observations and then it has to be the u_h the probability of one two right so u_h that arrives us at the conclusion that we have to maximize the model evidence P_{FO} so maximizing P of O is also minimizing equal to minimizing minus $\log p$ of O right because u_m it is minimizing the surprise u_m so in the beginning we say that we continuously update our belief after making an observation of the environment u_m we call the belief after making an observation a posterior u_m this is Q of s right u_m so to compute posterior we want to put in a u_m we just

want to call this you know any kind of distributions over States then for example we can sample it we can randomize it right and then we try to gradually trick this distribution so that it's used the highest modal evidence or it's used the lowest surprise so how do we choose the actual posterior so that it has the lowest surprise right so in minus $\log P(O)$ here is the actual minimizing surprise that objective that we are wanting to minimize right so we can analyze it like we can do it with a base theorem this not Bayes theorem but base formula or theories yeah and then to evaluate the surprise minus $\log P(O)$ we need to sum out the hidden variable s from the Joint distribution $P(O, s)$ but well this summation there's a problem right because that we do not know like all the hidden States all the hidden variables in the environments it is hidden from us right so it is normally very very hard to sum up all these State values so uh if we cannot sum it up then we can try to approximate it so first we introduce a posterior $Q(s)$ into this sum right we multiply both the denominator and the numerator by the posterior so this is the belief that we are trying to tweak right so in here we're trying to take this posterior so that we can minimize this term so this gives us a ratio $\sum_s P(O, s) Q(s)$ where s is the ratio $P(O, s)$ and $Q(s)$ over the distributions $Q(s)$ is weighted by the $Q(s)$ this is like weighted sum right so according to the inequality of convexity or Jensen inequality says that's the convex function of a weighted means is always smaller than the weighted means of the same convex function so you can see the convex function here is minus log of something right so minus log of something is just a hyper uh is just a parabol something like here and uh this is smaller than a weighted means of the complex functions also we put this some outside right so this is also applied to the context of probability Theory and then we can arrive at this uh yeah this equations so formally it's called an upper bounds uh this one right a quantity that approach from the above right so this is also means that this is a free energy um when we find a posterior that minimize this free energy term we will have a chance of approaching the lower surprise points um which potentially increasing the modal evidence so after we have this terms right here we can just put the minus inside and then it's become this form this is the conventional form that we call variation of free energy um and after decomposing this we will have these two form uh which is the Kullback Divergence of between the posterior and the prior uh minus the weighted sum of the log of $P(O|s)$ the $P(O|s)$ or given s which is the posterior functions or no this is likelihood so basically the Kullback Divergence term is called the complexity and the minus $\log P(O|s)$ is called accuracy term where the complexity terms compute how supplies are State to State distributions posterior and the prior are

far away from each other or not right and the accuracy terms uh defines how accurate our model is predicting um the actual observations right so this is also relatable to the variational auto encoder where um we compute the loss of the variation Auto encod um to be the K Divergence of the posterior and the Gin distribution with a mean of zero and a standard deviation of one and then we also have the uh mean Square arrows of the actual predictions of the images right um and then VIP and MF is a little bit um different than from variation of out coder VP is variation of free energy and mfp is marginal free energy as in the um active inference tutorial um the paper U so marginal free energy says that um we compute the CH Divergence between the poster and the prior and the prior here is computed from the transition model um not the um not the normal distributions of mean of zero and this standard equation of one so uh here is the uh General model of trying to build a deactive inference model right so here's first of all we're trying to encode right using the posterior functions so in active inference this is the p s given o is the transpose of the likelihood Matrix P of given s this is normally learns with the cical distributions and using the orally conjugate prior um and then uh in the Deep learning literature um this is often known as the encoder U that's uh Often parameterized by the neuron Network um and then o here is often be an image and state here is just um some hidden vectors that we are trying to compute and then next one is the um La function right so normally we text in the states um of um the agent and then we're trying to estimate the uh predictions the observation predictions right um so in traditional active INF this is known as the likelihood Matrix or likelihoods um per se um this is also learns with the Cal categorical distribution and using the conjugate prior and deep learning literature um this is known often known as the decoder um that is often parameterized by a Neo Network so uh as we know the variation of free energy formula the complexity terms K Divergence here right and the um uh accuracy term is the minus $\ln p$ right given s um so here we compute the K Divergence between the encode States and the actual prior States here and then um next one we comput the um actual L probability of the predicted observations and the actual observations right here um and that makes the um whole objectives where we compute the two uh objectives right here that's uh also be the um variation of free energy formula so we just talk about how inference um inactive instrument work but there's more we want to build an inference system that can interact with the world right so when it comes to interaction between the agent and and the environment we also need to have action and um also roll out in time step right so basically we have another elements here which is at and remember I also mentioned the transition model right so after we take an action in the world um we um the the

world u^m or the environments evaluates one step right and it uh it change its own State and then emits a new observations u^m and with we as agents or intelligence system also try to predict u^m given the action in the current states what is the next States and what and this States is the prior over u^m the states of the next town step where we have not made the observation O^t plus 1 yet so after we have the S^t plus 1 as prior u^m we do the same process as I have mentioned before so uh we make up observations u^m try to change u^m our belief uh after making that observations and then learns from the uh new observations to take our prior into the posterior so here we learn it u^m using the variation of free energy or marginal free Energy Formula so as we can see from the previous slides uh we do need somehow U to somehow plan the actions to interact with the world so if we're trying to minimize or surprise u^m with the current observations u^m so in the action planning we aim to take actions that minimize the similar free free energy term in the future u^m this is often called that's the expected free energy by the way u^m so in here uh we introduce some new terms right Θ u^m is the model parameters u^m the reason why this is here is because we want to also compute the u^m uh modal the surprise over the modal parameters or the uncertainty of the modal parameters u^m so you can see that the expected free energy in the future it's kind of similar to the version of free Energy Formula it's only needs u^m another model of parameters parameters and the given the p_i or the actual plant's policy or plant actions so after a lot of derivations right I'm we're not getting into like the real deviation here but u^m here's the end results that's u^m something something minus u^m minus the weighted means of the uh L P of O^t right so I'll just focus on this u^m term is because this is the u^m goal of the talk today so basically u^m this term lock of p is the expected free energy derivations on of the instrum uh instrument mental term so basically uh this also means that we are trying to maximize the now we're trying to take the action that maximize u^m the future observations that matches our u^m prior parap distributions or what we want right so normally u^m to learn to take actions based u^m uh to learn to take actions u^m to align with what we want to see in the future u^m we of needs to craft and provide the priate preference manually in advance u^m and then we T the option based u^m we take the options u^m that's maximize the likelihoods u^m given that distributions u^m but this is often not really practical in the realistic World u^m H because we cannot like craft the manual distributions like every single time we encounter new environments or encounter a new task right so what we can also do is we can also provide a goal Stat or goal observations directly to the agents which is more practical u^m but in the end the prior preference distribution is very sparse u^m and this provides little to no learning signal to the

agent as you can see on the figure on the right and the left right so left is a prior preference distributions and right is a single goal distribution or goal state image so uh what can you do to dynamically uh produce um the uh uh prior preference right we can try to learn a prior preference model right that um produce a goal image at each time step so here is on top is the prior preference model that predictions right and uh in the bottom here is the model prediction of the future um given the sequence of planned actions so we are trying to say that we um we want to take the actions um that have uh the roll out in the futures predictions matching the prior preference uh roll out in the future so how do we do that we first have to learn the model first right so in learning uh we can try and say that each time we achieve a goal then then we Mark the whole trajectory as the prior preference trajectory or the um positive um trajectory or expert trajectory and then for each observations marks as go we use a deep learning like another deep learning model to estimate start observations um for example right and then um in inference time um we actually train when we train the policy planner uh to propose the actions right we make the policy planner proposing the actions first and then predicts the future observation based off of that um actions U that is planned by the policy planner and then um we actually learn the policy planner based on the signal that is emitted from how aligns the predicted future observation would be um to the predicted goal um the predict the predicted goal here is the actual prary props model outputs right um then we train the policy planner with the um mean Square Arrow or um some kind of objectives um that's uh predicts uh uh some kind of objectives between the predicted goal image and the predicted future observation so there's this problem uh where uh uh it exists between all these learning schemes because it is very computational expensive um so it is generally uh expensive because we have to work directly with the images so for example in learning we have to use a decoder of the um of the PRI preference model um to actually predict the goal image at each time step right um in inference uh time is also computationally expensive is because um it has to learn from the images itself and computed down from uh the um the back propagations will have to flow through the decoder back to the action planner so here it comes to the actual methodologies uh we actually use Dynamic PR preference learning um in the state space um instead of the observation space right so first of all I want to talk a little bit about the world model so uh this uh model architecture is um basically what I have just um drawn in the background sections but this looks with robots images so in here we have the observations um O_t O_t and here's also the observation the same observations O_T minus one O_T um and we text in the encoder and is the encoder posterior texting the encoder and we

have the transition prior μ and then we have the decoder right encoder encodes the image in
 have the states and then decode the μ the image the state back to the image and then
 computes the accuracy loss right accuracy loss $\mathcal{L}$ is minus $\mathcal{L}$ prop of the P given S as we have
 discussed in the variation of pre energy sections and then the complexity is the k k Divergence
 term between the posterior and the prior right so how do we compute the prior we compute
 from μ the previous steps μ States and then we use the transition model μ to estimate the
 next States prior right so that's μ that's how we can get the μ prior so μ current work μ
 and popular Works μ regarding the world model or generative models μ often in μ the
 model called μ RSS language stands for μ recurrent State space model μ in the translation
 μ model so it's to zoom out a little bit μ we are trying to μ integrate some kind of temporal
 informations Through Time right μ so that's the state of the environment or the states that
 we're trying to predict be more accurate μ and it carries more informations with time μ so
 for example μ we cannot μ encode the μ enough informations from just a single image
 because for example μ there can be like μ position and velocity right the velocity is
 acquired μ through multiple images μ and cannot be acquired from a single image if does
 Mak sense so what we of to do is we try to integrate a recurrent NE networks that text in the
 previous States μ or the previous recurrent States H my T minus one μ and then we also
 text in the μ encoded observations μ and then with in transition μ we produce the
 recurrent neur network will produce the current recurrent States μ and also predicts the
 outputs to to put into the μ transition model μ and also it's the recurrent stat will also μ be
 served as inputs to the posterior Network μ so that's basically μ the idea and then μ the
 other way μ around we could just μ do μ the model as normal so μ in active influence asan
 texting the actions μ that's it belief μ that would lead to its preferred outcome μ using
 instrumental signal μ to construct this prior preference μ Pas has μ for example provide a
 go statement observations directly to the agents for example mazaka in 2021 with contrastive
 active INF μ we can also manually craft a PR prence distributions μ we can also
 dynamically produces the preferred observation that each time set μ but as I mentioned
 already μ all these method is one way or or another very expensive μ or it's very not
 practical or in the reward applications or μ it is very computationally expensive for
 dynamically producing the actual observations μ for the the goals so we can also use
 Behavior cloning but it's its trajectory diverges from on the trading trajectory due to μ small
 mistakes made by agent as well as environmental μ stochasticity μ and also it is because of
 ident IND independent and identically distributed assumptions of positive exper data

distribution so we can um we we try we will kind of point it because it's it's not going to um be uh the same right the two distributions of the actual um actual trajectories in the environment will not be the same as um every single positive trajectory of the expert data so to tackle this we try to leverage a small quantity of seat imitation data to learn an uh a neur network that's dynamically produced preference Over States not observation but State um at each time step so this effectively um produces and easily generated dense instrumental go signal um and we also design the agent such as that it moves according to the trajectory that is is shaped toward its own estimation of the future States or future preferred States so it's trying to notch the its own trajectory on toward watches the imitation positive data distribution so as you can see on the figure on the left um so you can see that in Behavior cloning it's trying uh the actual trajectory is diverges away from um the actual imitation data where the prior preference over the states trying to um get uh trying to be shied toward the imitation data itself using the model signal and ra is our agent um is called robust active inance um and it learns um from both epistemic signal and instrumental signals but in here we'll just focus on the instrumental signals and how to construct it so uh yeah so basically we use the decoder um or the likelihood functions of the generative model to only visualize on the preferred observations um not that in training or test time we do not use the motor learning uh we do not use the actual observation uh goal observations by itself we only use the hidden States so the goal here is to um get the states the goal States at each time step be predicted by the prior preference model right um You can see that's uh on the top here is the real trajectory and the bottom here is the actual um dream that's the dream of the robots that it wants to be right um and then we have predicting that dream right so here is the actual model of the Contra recur and state prior preference learning um uh so basically it has um it has also similar architecture to the rssm to the world model um it also has have an encoder um here it also have translation um but it does not have the decoder um so how do we actually uh learn this uh um prior preference model so first of all uh we uh learn the world generative World model as normal right um but when we say whenever um it's uh whenever the trajectory is uh positive or a successful trajectory um we try to contract um the state of the uh crispy I just call it crispy um we try to contuct the state of crispy to the um world model States and then um whenever the um trajectory is negative we try to push away the states um from each other so that the crispy model um can only predict the um positive States so how do we determin um if it is a POS trajectory um or the negative trajectory without having a very sparse signal so we will have a thing called brow or the prior preference signal so how do we compute this raw signal so imagine um this is the

real trajectory um and it is a successful trajectory for example right so we're trying to compute
 um the prior preference states by having the rate of the successful States is at one and it starts
 from one and it it be case backward so the case um very fast and um if so in this like really far
 away States from here um they do not have um enough signal because there's no goal um like
 there's no direct go from that states and then there's real trajectory that is failed right um You
 can see that the mountain car cannot reach um the fls on the right here so whenever it's built
 um we can just PL plat the very very last observation as minus one and then we start to De
 Decay backward negatively um and then uh we have the prior preference rate signal as um
 very negative so the reason why it is it has to be last is because um we want to um we want to
 blame the L observations um hardest so um in order to uh in order for the agents to just push
 away from the failing States very very very strong um in the very last stage so in the real
 trajectory that have like multiple successes um we try to put all the one in all the successful
 States and then just Decay backward um as similar to how we Decay backward in a single
 success case so with that s we now uh know how to comput um the RO and then this will
 serve as the dynamic signal um to the contrasted loss and it creates a dynamic behavior of the
 contrasted loss um that's trying to either push away or um contract um the two states of the
 preseed state and the world model States so for example in here um there's if there's positive
 row um the contracted block would be um Contra Contracting all the states together and
 maximizing the similarity between the precipitate states and the world model States and then
 we also learn to minimize the K diversions between the um complex uh uh the posterior and
 the prior uh of over the states of the crispy model um this is because we want to estimate the
 translations right so we translation to the um next goal State and then this goal state has to
 match the actual goal um that is the current stage of the envir of the rsm model or the world
 model um so in contrast right if row is negative then we're trying to push away um all these
 states away right so we're trying to maximize this um distance or minimizing the similarity
 between the um world model States and the cryst spe State and in here notes that there's no
 complexity objective is because we don't want to um predict the next goal that is um the
 failing State that's basically the the idea so that arrives at the uh general formula for the
 objectives of the contrast State PR preference model um basically uh it is just masking out um
 The Sign of the row and then multiply with the the K Divergence or the complexity uh
 between the posterior and the prior of the crisby model um and then we uh minus the row and
 SE of um the actual predicted State and the world model state so same here can be very much
 um uh any kind of similarity functions for example um cosine similarity um and then in the in

our work we actually use the actual coin simulator so in training time we have we have just talked about training time right so we have successfully assuming that we have successfully learned the prior model that predicts the future goal and in inference time we've used this prior transition model to estimate future rollout trajectory of the goal right so we can see that on the top here is the observation actual observations and in the second row is the actual prior preference rollouts or predictions of the future goal uh given the current observation so given the prior actions we also train the plant uh the plant actions right so the actor trying to predict actions given the observations the predicted observation in the future so we're trying to say uh this is uh we're trying to compare uh the actual prior preference rollouts with the actual predictions given the actions and then we are trying to learn the action planner based on the signals or how divergent the future goal is to our future predictions of the world given the options so uh there can be so many reinforcement learning agents and active agents that can utilize this technique of Prior preference learning so active INF agents I put my works here it's rare and then there's also Chris Bakley's work and Baren M work about deep active influence in variational policy gradients methods and also Auto work is the scaling of the active influence in PDP and not in car environments and so on right so reinforcement learning agents there's all these sorts of actor-critic algorithms like dreamer series soft actor-critic dreaming ddpg tqc and so on so there's a lot of agents that can utilize the framework because there's no change in the original architecture there's just an add-on to how an agent would apply the private preference learning signal into the system so is the actual flow chart of the agents basically we start the agents and for each episode we evaluate the environments using the world model and the policy planner actually basically we evaluate in the environment we let the robots play we let the robots do the task and then we collect the data and add them to the replay buffer for example right and then we can get the main training agent logic and then first of all we sample the training data from the buffer so in here we're trying to have we also have to use the self-imitation Learning buffer so uh yeah so self-imitation learning is very good technique normally to balance the number of successful and failing trajectories and we also learned so first of all we learned the world model encoder-decoder transition on RSSM and then we train the epistemic model but in this case we don't need to care about it and then next is the Contra Rec St PR preference model from the data and then next we imagine the future trajectory using the world model and the policy planner right we

try to estimate the future options um to be um what is the optimized future actions for example which we estimates the future action based on how uh current state is and then the world model will be continuing to bouts in the Futures and then we dream of the futures and next is the part that we actually um infer on the uh crispy model we use a crispy model to imagine the future preference trajectory um given the goal um given the current observations and we predict the future goals roll out goals right so after that we learn the policy planner right using the instrumental signal the instrumental signals here is computed from the um difference um between the future goal and the um predicted future dream of the world right and then with that the policy planner can learn from uh from the uh instrumental signal so here's our um preliminary results on um active influence agent so basically um we can see um the two top rows is the mountain and car environments and the two bottom row is Rob robotics met a world um push button environment so let's look at the first row for example the observation here is a mountain paru have reached the goal and for each um observation here we uh we predict a preference roll out or preference um or a goal State at that single time step so in here we can see that um if the car has not have the moment yet um then we have to build the momentum by predicting the car going to the left for example right and then after it has built some kind of momentum we trying uh the actual prior preference model predicting the car would go to the right go forward right when it goes here and the prence model will predict the car to be um going closer to the goal something like this so the same applies to the um robot environments where the actual observation is the third uh row and the uh prior preference roll out is on the fourth row so we can see that uh it is also trying to estimates the dream or the dream of the goal of the robots is to actually push the button so um there's different kind of experiment setup that we are trying to learn um basically we have uh four trials on each environment and then we train the experts um using as sof critic or po proximal policy optimization um to collect um a small amount of seed data normally is about uh 20 episodes something um 20 to to 30 episodes um so we test mostly on three mains environments or suits um first of all is Mountain car there's only one task um metal W is 13 task so here's um all red robots here's metal war and then the robot suit we have two task and then um the C regarding camera setups uh we use um three views for met world and four views for um the robo suit so the green check mark here denotes uh that the robots has uh successfully solve the task um here's a robo suit door open door and then um here's Robo suit lift where it can successfully lift the block and here's more U solving agent tasks so basically agent with Crispy exhibits and ability to perform successfully early on um and uh so agent with without crispy uh struggle to

actually succeed on solving the environment um and we also test on another other environment as well for example gasum robotics or and risk task um you can you can see in the middle image on over here so actually you go to the discussion sections so actually um I want to make a note that um with respect to the crispy subm model um RM model utilize contrastive objectives to adapt its parameters so basically it solely learns on um estimating the world model in coted States while pushing itself away from undesired World model States or trajectory so this does not require or the learning of the of the decoder so we we only make sure of the Learned decoder in the generative World model um to visualize the uh to visualize the preferred observations um from the estimated crispy States so experimentally we remark that this is an um that's we only use um this decoder um to as an auxiliary tools um to prove um that the states of the contusive recurrence Prime preference model uh is useful so theoretically Behavior cloting will be unable to achieve a great uh success in multile multi goal environments due to its Reliance on a fix and finite size uh imitation sample pool um the expert trajectory in this fixed pool May further differ from the actual trajectories um that the agent needs to take to solve the problem at hand um for example there's a distributional gap in in observation and there and um it requires different sets of actions to be taken for example right um so on the other hand right crispy learns to produce dynamically um the goal sticks and not the go observation right um so we can empirically confirm in simulation that our agent um with Dynamic prior preference learning does not suffer from the goal miss much problem uh in the training phase so this is very applicable applicable to the multi-goal environments so um we can see that um this P preference learning uh techniques has a lot of practical applications in different robotic tasks um with for example those with potential dangers in the operations um and functioning for example when human safety must be considered right um for example a um trying to drive a autonomous driving car system where uh it has to have or it has to predict um the future um goal right the future goal is not to hit the human for example right um and we can also make a lot of advancement in imitation learning in research community and it can be um applied by the wide range of um reinforcement learning in active inference agents um and then the last one is to scale the the active inference agents on um the real world robotics agent so future work words actually considered um importing or integrating the dynamic PR preference learning in latent State space model um for example variation of Dynamics discret variable Auto encoders or tension waiting models um and also uh there's another models uh physical neur robotic system um which entails an embodied and active and survival oriented formulations of active perception and World model learning so in

this uh talk um uh we have discussed a lot of ideas about being able to estimate the particular goal or preferred State um at any particular time step is very useful um for cognitive control agents so as our results demonstrates the agent can then learn to adapt the tech actions that leads to um a estimated goal um and then um in contrast to the approach is taken in the imitation learning and behavior cloning literature um which train the agent's policy based solely based on the fixed collected expert data sets in our framework the expert signal is estimated dynamically um through an Adaptive prior preference model right closing this Gap the domain gap between the actual trajectories and the collected trading imitation trajectory or preferred trajectory um and then we also discussed the generative model um and and coder decod transations and how the variation of free energy work and then how to learn the model and then we finally we discussed the contrasted recurrent State prior preference model um with a self-revision mechanism and then discuss the model learning and then inference um object deted um and how to integrate stat into the agent planner learning system and then we show the results for um the robotic task and then I mention how it can be applied to different fields so that concludes my presentation um and thank you very much for um listening thank you that was amazing really great job with the background math done only like a PhD student can but it was great to see how you brought it all together and connected your research to it all at the end so thank you that's all for now see you later thank you see you e e